📘 AI/ML Internship – Day 37 Notes & Tasks

🟢 Module 7: Classification vs Regression in Machine Learning

📌 Part 1: Introduction

👉 In Machine Learning, problems are mainly divided into:

1. Regression
2. Classification

👉 These are two major types of Supervised Learning.

📌 Part 2: What is Regression?

🔹 Definition

👉 Regression predicts continuous numerical values.

👉 Output will be numbers.

🔹 Examples

- House price prediction
- Salary prediction
- Temperature prediction
- Sales forecasting

🔹 Example Dataset

Experience	Salary
1	20000
2	30000
3	40000

👉 Output is numeric.

📌 Part 3: What is Classification?

🔹 Definition

👉 Classification predicts categories/classes.

👉 Output will be labels or categories.

🔹 Examples

- Spam or Not Spam
- Disease or No Disease
- Pass or Fail
- Cat or Dog

🔹 Example Dataset

Study Hours	Result
1	Fail
4	Pass
6	Pass

👉 Output is category.

📌 Part 4: Theory Answers (IMPORTANT)

🔹 1. What is Regression?

👉 Regression predicts continuous numerical values.

🔹 2. What is Classification?

👉 Classification predicts categories or labels.

🔹 3. Difference Between Regression and Classification

Regression	Classification
Predicts numbers	Predicts categories
Continuous output	Discrete output
Example: salary	Example: spam detection

🔹 4. What is Continuous Data?

👉 Data that can contain any numeric value.

Example:

- Height
- Weight
- Salary

🔹 5. What is Categorical Data?

👉 Data divided into categories.

Example:

- Male/Female
- Pass/Fail
- Spam/Not Spam

📌 Part 5: Regression Example

🔹 Linear Regression

```
from sklearn.linear_model import LinearRegression
import pandas as pd

data = {
```

```
    "Hours": [1, 2, 3, 4, 5],
    "Marks": [20, 40, 60, 80, 100]
}

df = pd.DataFrame(data)

X = df[["Hours"]]
y = df["Marks"]

model = LinearRegression()

model.fit(X, y)

prediction = model.predict([[6]])

print(prediction)
```

👉 Predicts marks (number).

📌 Part 6: Classification Example

🔹 Logistic Regression

```
from sklearn.linear_model import LogisticRegression
import pandas as pd

data = {
    "Hours": [1, 2, 3, 4, 5],
    "Result": [0, 0, 0, 1, 1]
}

df = pd.DataFrame(data)

X = df[["Hours"]]
y = df["Result"]

model = LogisticRegression()
```

```
model.fit(X, y)

prediction = model.predict([[6]])

print(prediction)
```

👉 Output:

- 0 = Fail
- 1 = Pass

📌 Part 7: Visualization of Regression

Study Hours ↑
Marks ↑

1 → 20
2 → 40
3 → 60
4 → 80

👉 Continuous increase.

📌 Part 8: Visualization of Classification

Study Hours → Result

1 → Fail
2 → Fail
5 → Pass
6 → Pass

👉 Category prediction.

📌 Part 9: Understanding Output Types

Problem	Output
Salary prediction	Number
Weather prediction	Number
Spam detection	Category
Disease detection	Category

📌 Part 10: Real-World Thinking

🔹 Regression Applications

- Stock market prediction
- House prices
- Temperature forecasting

🔹 Classification Applications

- Face recognition
- Email spam filtering
- Fraud detection
- Medical diagnosis

📌 Part 11: Important ML Terms

Term	Meaning
Feature	Input data
Label	Output data
Regression	Numeric prediction
Classification	Category prediction

📌 Part 12: How to Identify Problem Type?

🔹 Ask this Question:

👉 "Is output numeric or category?"

✔️ Numeric → Regression
✔️ Category → Classification

📌 Part 13: Regression vs Classification Flow

Input Data
   ↓

Is output numeric?
   ↓ YES → Regression

Is output category?
   ↓ YES → Classification

📌 Part 14: Why This Topic is Important?

👉 Before building ML model:
Students must identify:

- Problem type
- Correct algorithm

👉 Wrong selection causes poor predictions.

🎯 Tasks for Day 37

🧠 Task 1: Theory (Write Answers)

1. What is regression?
2. What is classification?
3. Difference between regression and classification

4. What is continuous data?
5. What is categorical data?

💻 Task 2: Regression Practice

1. Create marks prediction dataset
2. Predict salary
3. Predict sales values

💻 Task 3: Classification Practice

1. Pass/fail prediction
2. Spam/not spam example
3. Disease prediction example

💻 Task 4: Identify Problem Type

1. House price prediction
2. Face recognition
3. Weather forecasting
4. Fraud detection
5. Student marks prediction

💻 Task 5: Dataset Analysis

1. Create regression dataset
2. Create classification dataset
3. Identify features & labels

💻 Task 6: ML Logic Thinking

1. Explain why salary prediction is regression
2. Explain why spam detection is classification
3. Compare output types